Procureline
PROCUREMENT SYSTEM

TEFOMA CONSTRUCTION · 15 BUSINESS UNITS

Project Scope Document

Integrated Procurement & Stores Management System

What the system does, who does what, when, where, why and how

CLIENT
Tefoma Construction

BUSINESS UNITS
15: 12 in Zimbabwe, 3 in South Africa

VERSION
2.0

DATE
5 November 2026

BASIS
Code on the main branch, 5 November 2026

CLASSIFICATION
Internal — confidential

Contents

01 Document control 3
- Version history 3
- Review and sign-off 3
- Related documents 3

02 Executive summary 4
- What it guarantees 4
- Where it stands (5 November 2026) 5

03 Why: background and objectives 5
- The problems it was built to solve 5
- Business objectives 6

04 What: scope 6
- Organisational scope 7
- In scope 7
- Out of scope (not built) 7

05 Who: roles and responsibilities 8

06 When: the end-to-end process 10
- Time limits and schedules 11

07 Approval matrix 12
- Purchase order approval chain (since 28 October 2026) 12
- Other approvals 13

08 What: functional modules 13

09 Supplier lifecycle 16
- Two-person activation 16
- Due diligence by tier 17
- Ongoing controls 17

10 Where and how: architecture and technology 19
- Technology 19

11 Security, audit trail and notifications 20
- Access control 20
- Audit trail 21
- Data retention 21
- Notifications 21

12 Compliance, rollout, assumptions and risks 22
- Compliance with the procurement procedure 22
- Rollout plan 23
- Assumptions 23
- Constraints 23
- Risks and open items 23

13 Glossary 24

SECTION 01

Document control

This document sets out the full scope of Procureline as it is built today: what it does, who uses it, when each step happens, where it runs, why it exists and how it works. It describes the system from its code, not from a plan. Where older documents disagree with it, this one reflects the current build.

Audience: executive sponsors, business-unit management, Procurement, Finance, Stores, auditors, testers and the technical team taking over support.

Version history

Version	Date	Change
1.0	4 Nov 2026	First full scope document. Covers the three-stage purchase order approval chain introduced on 28 October 2026 and the Procureline brand from 29 October 2026.
2.0	5 Nov 2026	Supplier activation now takes two people: Procurement submits, the Head of Procurement or the COO approves, returns or rejects (sections 5, 7 and 9, figure 2). Registrations can be rejected and reopened; a company cannot register twice; permanent supplier codes; business and entity type. Invoice matching now accounts for earlier invoices on the same order, and an expired required document blocks awards on its own. A department head's own requisition needs no separate approval; store requisitions only collect delivered goods. Suppliers are told of every RFQ that opens, every GRV and can download their order PDF. The hourly jobs need an external scheduler on the hosted API (sections 6, 10 and 12).

Review and sign-off

Role	Name	Signature	Date
Prepared by			
Reviewed by			
Approved by			

Related documents

Document (in the code repository)	What it holds
`BUSINESS_REQUIREMENTS_DOCUMENT.md`	Business requirements and acceptance criteria
`PROCEDURE_COMPLIANCE.md`	Clause-by-clause mapping to the Central Procurement Procedure (Rev 9)
`EXECUTIVE_BRIEFING.md`	One-sitting summary for senior management
`docs/SOURCELINE_SYSTEM_GUIDE.md`	Technical and operational guide

Document (in the code repository)	What it holds
`docs/ENVIRONMENTS.md`	Development, staging and production set-up
`docs/MULTI_ENTITY_PLAN.md`	How the 15 business units are separated
`docs/DATA_PROTECTION_ASSESSMENT.md` `docs/DATA_RETENTION_POLICY.md`	Personal data handling and retention
`docs/AUDIT_EVIDENCE_PACK.md`	Evidence of controls for auditors
`docs/SOURCELINE_ROLLOUT_PLAN.md`	Infrastructure handover and the phased go-live
`docs/Tefoma-Full-System-E2E-Handover-Test-Pack.html`	End-to-end acceptance test pack

> Some of these still describe a four-stage purchase order approval with the department head first (removed 28 October 2026) and single-person supplier activation (replaced 5 November 2026). Sections 7 and 9 show the current rules.

SECTION 02

Executive summary

Procureline is a web-based procure-to-pay and stores platform. It runs each purchase from the moment someone asks for something to the moment the supplier is paid, in one system with a full audit trail. It serves 15 business units, each on its own web address with its own database.

It replaces paper requisitions, emailed quotes and spreadsheet tracking. It also turns Tefoma's Central Procurement Procedure (Rev 9) into rules the software enforces, not guidance people can skip.

What it guarantees

Three quotations before an award

Fewer is allowed only with a recorded waiver and a reason from a fixed list.

The right approvals, in order

A purchase order is approved by the Head of Procurement, then Finance, and also the COO at USD 5,000 or more.

No payment without receipt

An invoice is matched against the order and the goods received before Finance can approve it.

Only vetted suppliers are paid

A supplier can quote while its Know Your Supplier (KYS) checks run, but cannot win an award or receive an order until they are complete and a second person has approved its activation.

Everything is on record

Every create, approval, rejection, upload and sign-in is logged with the user, the time and the IP address.

Each unit stands alone

Every business unit has its own address and database. A session in one unit is refused by all others.

Where it stands (5 November 2026)

Item	Status
Deployment	Live. All 15 business units serve over HTTPS on their own hostnames.
End-to-end proof	The full cycle (50 steps) ran against the live system on 26 October 2026 with no failures. The scripted cycle now also covers two-person supplier activation and is due a re-run before the next go-live wave.
Data separation	Business units cannot see each other's data; a script check proves it.
Users	8 roles, including external suppliers through their own portal.
Size	About 69,000 lines of TypeScript (API, client and shared package); 26 data models, about 185 API endpoints, about 50 screens.
Before full use	Set-up, not software: load 14 empty units, enable real email for administrators, confirm the database backup tier, schedule the hourly jobs on the hosted API, and give existing suppliers their permanent codes.

SECTION 03

Why: background and objectives

Procurement ran on paper and email, so the controls in the written procedure were easy to work around. Procureline puts those controls in the path of the work, so the system refuses a step that breaks them.

The problems it was built to solve

#	Problem before	What the system does now
1	No visibility: nobody knew where a requisition was without phoning around	Every request has a number, a status, an owner and a full history, visible in real time
2	Duplicate buying: stores was skipped and items already held were bought again	Every approved requisition is checked against stock before Procurement can source it
3	Weak sourcing: emailed quotes could be altered, lost or arrive after the decision	Suppliers quote inside the system; each version is locked and time-stamped
4	Approval authority depended on who signed a paper, often out of order	The approval sequence is enforced in code and cannot be skipped or reordered

#	Problem before	What the system does now
5	Invoices were paid without checking what was ordered or received	A three-way match of order, goods received voucher (GRV) and invoice gates payment
6	Audits meant searching physical files	Every material action is logged with who, when and from where
7	One officer could register, vet and activate a supplier alone, and so decide who gets paid	Activation takes two people: Procurement submits, and the Head of Procurement or the COO approves; the submitter can never approve their own

Business objectives

ID	Objective	Measure of success
BO-1	Trace every purchase end to end	Any request is found by number, with status and history, in seconds
BO-2	Stop buying what is already held	Every approved requisition passes a mandatory stores check
BO-3	Enforce competitive, evidenced sourcing	Three quotations per award, or a recorded waiver with a reason
BO-4	Enforce delegated financial authority	Approvals run in a fixed order; the COO limit applies automatically
BO-5	Prevent improper payment	No invoice without an order; no approval without receipt and a three-way match
BO-6	Trade only with vetted suppliers	No award or purchase order until the supplier's KYS is complete and a second person has approved its activation
BO-7	Produce an audit trail without extra effort	Every material action is logged with actor, time and network address
BO-8	Encode the Rev 9 procedure in software	Policy rules work as system constraints, not guidance
BO-9	Serve the whole group with full separation	Each of the 15 business units has its own address and database

SECTION 04

What: scope

The scope is the full procure-to-pay cycle and stores for 15 business units, used by staff and external suppliers through a web browser. Each unit runs on its own and shares no data with the others.

Organisational scope

Country	Currency	Business units
Zimbabwe	USD	Fossil Contracting, Dokuma, Khayah Cement, Kurima Centre, Mandfar, Masimba, Persimmon, ProPlastics, Rhopower, The Grain Hub, Titan, Trends
South Africa	ZAR	Enviro Plastic, Mangethe, Thando Kinetics

In scope

Area	What is included
Requisitions	Raising by staff or by a department head for their own department, department approval with a budget check, stock check, acceptance by Procurement, cancellation with a reason, collection of delivered goods from stores
Stores	Item catalogue, stock by site, issues against requisitions, transfers between sites, reorder levels, low-stock alerts, bulk stock import
Sourcing	Requests for quotation (RFQs), supplier invitation by category, notification of every invited supplier when an RFQ opens, sealed quotations with brand and notes per line, ranking, award with justification, waivers, line-by-line award
Purchase orders	Raising from the accepted quotation, value-based approval chain, issue to supplier, supplier acknowledgement, printed PO
Receiving	Goods received vouchers (GRVs), acceptance or rejection, hazardous-goods MSDS, witnesses, faulty-goods return deadline
Invoicing and payment	Supplier invoices, three-way match that counts earlier invoices on the same order, VAT check, variance override with reason, payment recording, payment-run dates
Suppliers	Self-registration with a once-only check on the registration number, permanent supplier codes, business and entity type, KYS due diligence by tier, two-person activation, registration rejection and reopening, document expiry, vetting and site visits, bank-detail change control, evaluation, status (pending, awaiting approval, active, suspended, dormant, blacklisted, rejected)
Budgets	Annual budget per department, committed and spent amounts
Oversight	Role dashboards, approvals queue, reports, supplier reports, notifications, full audit log, auditor evidence pack
Administration	Users, departments (with head and deputy head), sites, business-unit settings, exchange rates
Multi-entity	15 business units, each with its own hostname, database, document numbers, branding and COO limit

Out of scope (not built)

Not included	Effect today
Group-wide reporting across units	No single view of spend across the group
A shared supplier portal across units	A supplier is onboarded separately in each unit it trades with

Not included	Effect today
Bank or ERP integration	Payments are recorded in the system; money moves through the bank outside it
Per-unit links in emails	Every email links to one fixed web address
Native mobile apps	The web app works in a mobile browser instead

SECTION 05

Who: roles and responsibilities

Eight roles use the system, and each sees and does only its own work. No single person can complete a purchase alone: the requester cannot approve, the approver cannot receive, and the receiver cannot pay.

Role	Who holds it	What they do in the system	What they cannot do
End user	Staff in any department or site	Raise requisitions for their department and track them to delivery	Approve anything, or see other people's requests
Department head (HOD)	Head of each department; a deputy can act for them	Approve or reject their team's requisitions; raise their own, which go straight to stores because submitting is the department approval; follow the resulting orders	Approve a requisition they raised themselves; approve purchase orders (removed 28 Oct 2026); select quotation winners
Head of Procurement (HoP)	The department head of the Procurement department (the Users screen labels them so)	Authorise awards; first approval on every purchase order; approve quotation waivers; approve, return or reject supplier activations; reject or reopen registrations; procurement cancel rights	Select a winner and authorise the same award alone; approve an activation they submitted themselves
Procurement officer	Procurement staff	Create suppliers and check their KYS; submit them for activation; issue RFQs; select the winning quotation with a justification; raise and issue purchase orders; handle bank-detail changes; evaluate suppliers	Activate a supplier alone; authorise their own award; approve purchase orders
Stores officer	Stores and warehouse staff	Check stock against requisitions; issue stock; receive goods and raise GRVs; accept or reject deliveries; transfer stock between sites	Approve orders or invoices
Finance manager	Finance team	Second approval on purchase orders; set department budgets; approve invoices after the three-way match; record payments	Raise orders or receive goods

Role	Who holds it	What they do in the system	What they cannot do
Chief Operating Officer (COO)	COO of the business unit	Final approval on purchase orders at or above the unit's limit (default USD 5,000); approve quotation waivers; approve, return or reject supplier activations; reject or reopen registrations	Raise or receive anything; edit supplier records
Supplier	External companies	Register once; upload and view KYS documents; quote on RFQs; acknowledge orders and download the order PDF; follow each GRV; submit invoices and see their payment history; request bank-detail changes	See internal records, internal approvals or other suppliers' bids
System administrator	IT administrator for the unit	Manage users, departments, sites, unit settings and exchange rates; read the audit log; cancel records	Perform business steps such as approvals, awards, receiving or payments

Four-eyes controls. A supplier bank-detail change needs a call-back to a phone number that was on file before the change was requested, and the approver must be neither the requester nor the person who made the call-back. An award needs one person to select it and a different person, the Head of Procurement, to authorise it. A supplier activation needs one person to submit it and a different person, the Head of Procurement or the COO, to approve it.

SECTION 06

When: the end-to-end process

A purchase moves through 20 steps in five phases, and seven of them are controls the system enforces. Each step unlocks only when the one before it is complete.

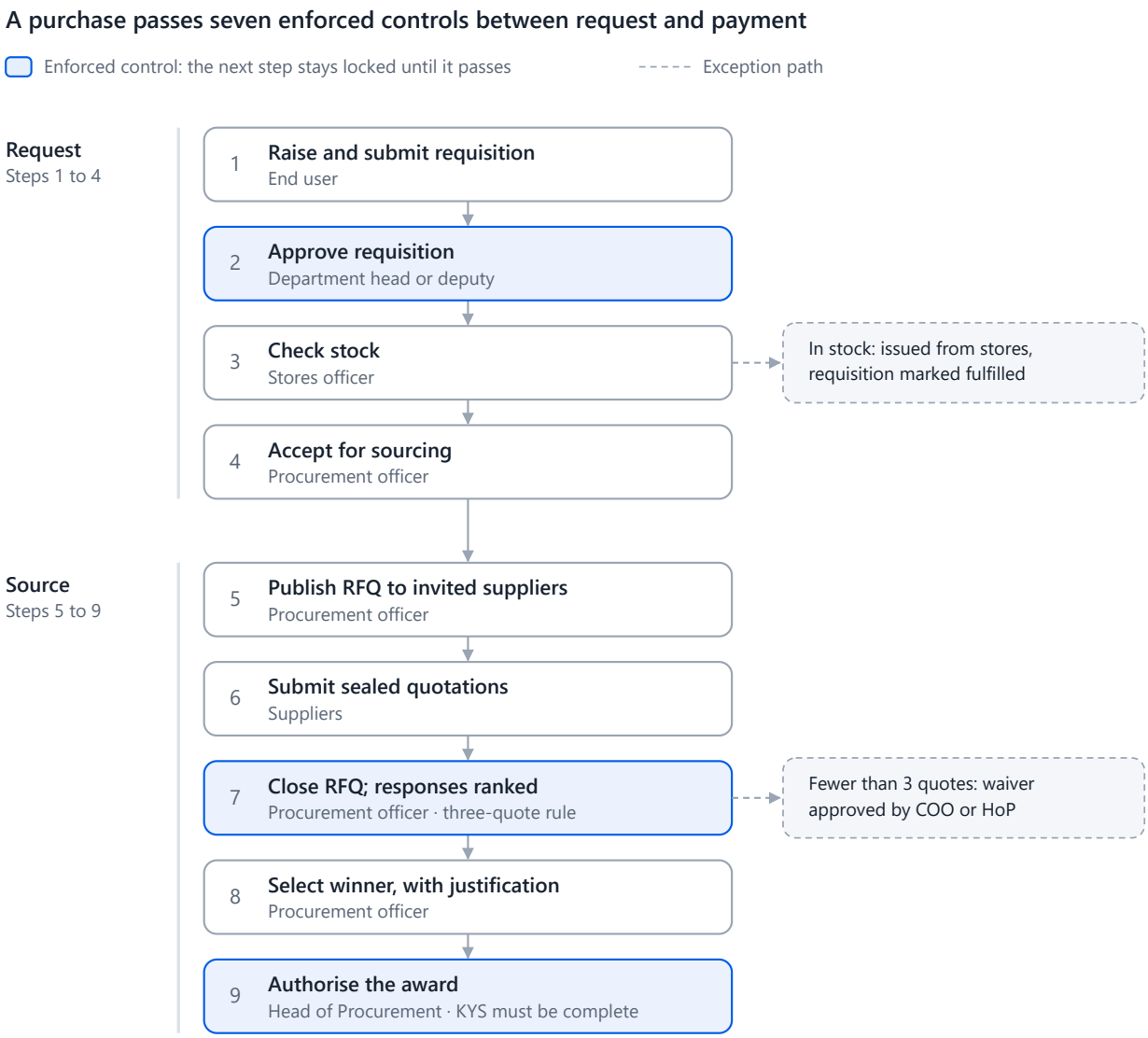


Figure 1a. Request and source: steps 1 to 9. Tinted steps are enforced controls; dashed boxes are exception paths.

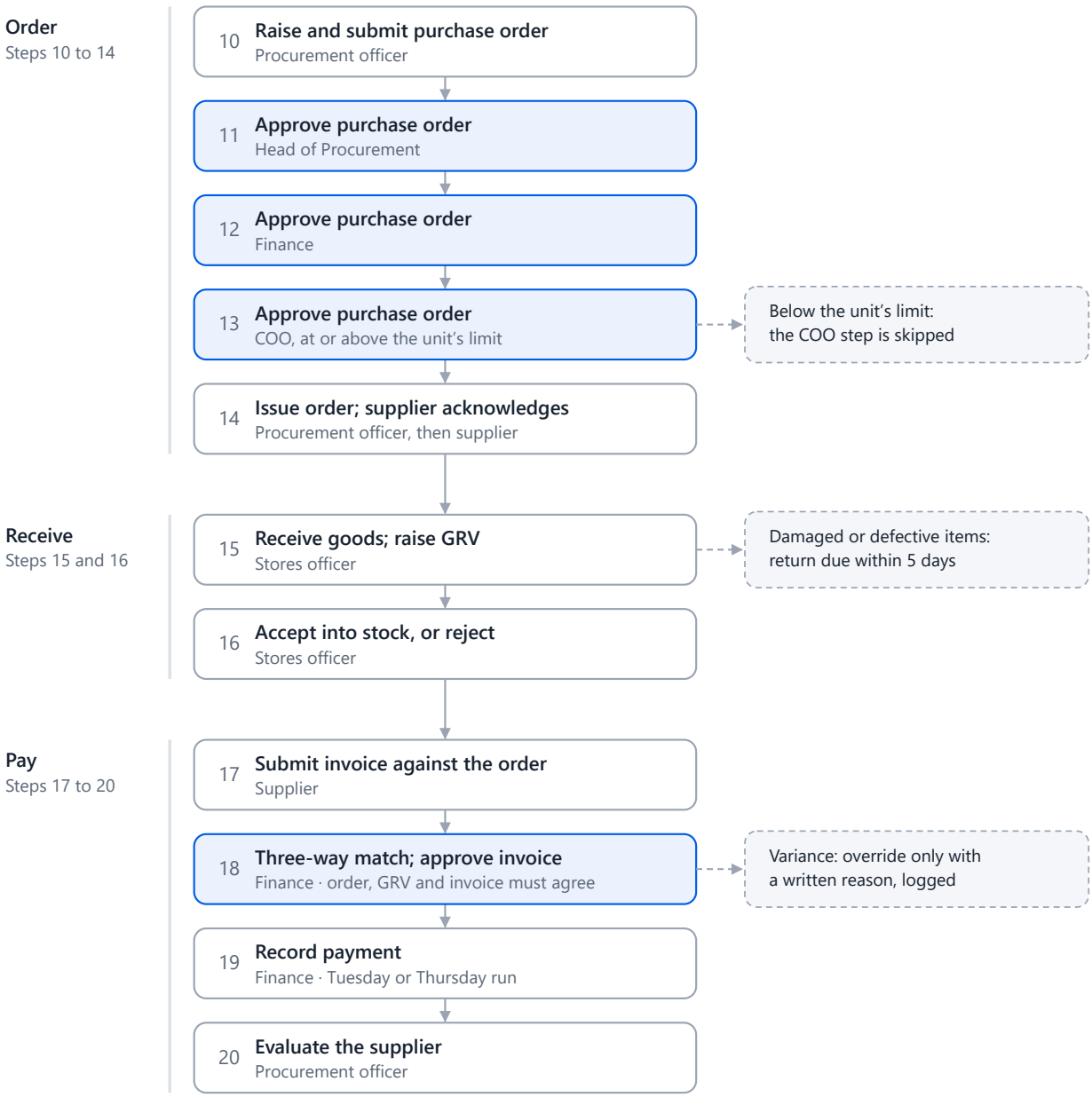


Figure 1b. Order, receive and pay: steps 10 to 20.

Any approver can reject with a reason, which sends the record back with that reason on file. A requisition, RFQ or purchase order can also be cancelled with a reason from a fixed list.

Time limits and schedules

What	Time limit or schedule	Who is alerted
Approving a requisition or order	1 hour (procedure target); the Approvals page marks items overdue	The approver
Sourcing an accepted requisition	2 days; the RFQ list marks overdue RFQs	Procurement officer
RFQ deadline	Alerts in the 48 hours before it closes	Invited suppliers

What	Time limit or schedule	Who is alerted
RFQ closes with fewer than 3 quotes	At closing, so the waiver does not sit unnoticed	COO and Head of Procurement
Faulty goods	Return deadline set to 5 days after delivery on the GRV	Stores officer
Supplier payments	Tuesday and Thursday payment runs	Finance
Supplier documents	Reminders 60, 30 and 7 days before expiry	Supplier and Procurement
Supplier re-evaluation	Every 3 months (critical), 6 (strategic), 12 (tactical) or 24 (transactional)	Procurement
Low stock	When stock falls to the item's reorder level	Stores

Background jobs check for these alerts every hour and run the data-retention job once a day. On a long-running server they run on an internal timer. The hosted API runs as serverless functions, which have no timer, so there they run only when an external scheduler calls the secured jobs endpoint; see sections 10 and 12. Every invited supplier is told when an RFQ opens, whichever way it was opened: published from a draft, created open, or raised to re-source unawarded lines.

SECTION 07

Approval matrix

Every approval belongs to one named role, runs in a fixed order and cannot be skipped. Purchase orders need two or three approvals, depending on the order's value in US dollars.

Purchase order approval chain (since 28 October 2026)

Stage	Approver	Required when	Status while waiting
1	**Head of Procurement**	Always	Pending Head of Procurement
2	**Finance manager**	Always	Pending Finance
3	**COO**	Order total is at or above the unit's COO limit (default USD 5,000)	Pending COO

- **Limit per unit.** Each business unit has its own COO limit in US dollars, set by the administrator. The default is USD 5,000.
- **Other currencies.** An order in another currency, such as ZAR, is converted to US dollars at the stored exchange rate before the limit is applied. The rate and limit used are saved on the order.
- **No exchange rate on file.** The order goes to the COO, so an unknown conversion can never skip the highest approval.

- **Older orders.** Orders submitted before 28 October 2026 may still show the old department-head stage. The Head of Procurement can approve them from there.

Other approvals

Record	Who decides	Rule enforced
Requisition	Department head or deputy	Checked against the department budget. Approving over budget needs a written reason, and Finance is notified. A head's own requisition is approved by submitting it; an over-budget one is reported to Finance as an override
Stock check	Stores officer	Items in stock are issued; only the shortfall goes on to Procurement
Award selection	Procurement officer	Needs a written justification, for the whole quotation or line by line
Award authorisation	Head of Procurement	Must be a different person from the selector; the supplier's KYS must be complete
Fewer than 3 quotations	COO or Head of Procurement	A waiver with a reason from a fixed list (single source, approved contract, no quotes received, unique product and others)
Invoice	Finance manager	Three-way match within 2% or 1 unit of currency against the goods received and not yet invoiced on that order; the approved invoices on an order can never exceed the value received; a larger variance needs a written override reason
Supplier bank-detail change	Procurement	Call-back to a phone number on file before the change was requested; contact numbers are frozen while a change is open; the approver is neither the requester nor the caller
Supplier activation	Submitted by Procurement; approved by the Head of Procurement or the COO	Two people: the approver is never the submitter. KYS for the supplier's tier must be complete, or the submission carries a KYS override reason that the approver accepts. The approver can instead return it with a reason
Supplier registration	Head of Procurement or COO	A pending registration can be rejected with a reason, and a rejected one reopened with a reason. A rejected supplier cannot quote, be awarded or be ordered from

Rejecting at any stage needs a reason, which is kept on the record and in the audit log.

SECTION 08

What: functional modules

The system has 12 modules across about 50 screens. Each screen shows only what the signed-in role may see and act on.

Module	Main screens	What it does	Key rules
Dashboard and approvals	Dashboard, Approvals, Authorizations	Shows each role its pipeline, counts and the items waiting on it	Only items at the user's own step appear; overdue items are marked against the 1-hour target
Requisitions	Requisitions, Requisition detail	Raise a request from the item catalogue (with package type and unit), submit it, track it to delivery, collect delivered goods from stores, cancel with a reason	Only the owner submits a draft; department heads see Team Requisitions (to approve) and My Requisitions (their own)
Stores	Stores review, Store requisitions, Inventory, Stock movements, Stock transfers	Check stock, issue items, hand over delivered goods, move stock between sites, import stock in bulk	Shortfall lines go to Procurement; an issue is all or nothing, checked before any stock moves; store requisitions only collect goods delivered against a requisition; stock below reorder level raises an alert
Sourcing	RFQs, Create RFQ, Quotations, Award	Invite suppliers, collect sealed quotations, rank responses, select and authorise the award	Suppliers are invited by category, topped up to at least 3; bids stay sealed until the RFQ closes
Purchase orders	Purchase orders, PO detail	Raise an order from an accepted quotation, route it for approval, issue it, print it	Approval chain in section 7; one order per accepted quotation
Receiving	Deliveries	Record goods received on a GRV; accept or reject each line	Hazardous goods need an MSDS reference; witnesses recorded; 5-day return deadline for faulty goods; the supplier is told what was accepted and rejected on every GRV
Invoices and payments	Invoices, Payments	Supplier invoices against an order, three-way match, approval, payment recording	No invoice without an order; no approval without receipt; payments run on Tuesdays and Thursdays
Suppliers	Suppliers, Supplier detail, KYS, Verification hub, Bank changes	Create, vet and submit suppliers for activation in a three-step KYS wizard; approve, return or reject them; suspend, blacklist or reactivate; manage documents and bank details; filter by Awaiting approval or Awaiting my approval; search by name, code or registration number	Activation takes two people; section 9 sets out the full lifecycle

Module	Main screens	What it does	Key rules
Supplier portal	My RFQs, My quotations, My purchase orders, My deliveries, My invoices, My KYS, Company profile	Suppliers quote line by line, acknowledge orders and download the order PDF, track deliveries, invoice and see payment history, view their own documents	Separate sign-in route; suppliers see their own orders and acknowledgement only, never internal approvals, match workings or other suppliers' bids
Budgets	Budgets	Set annual budgets per department; show committed and spent amounts	Finance sets allocations; approving over budget needs a reason
Reports	Reports, Supplier reports, Performance, Compliance, Evaluations	Spend trends, department spend, order values, supplier registry, compliance and paged evaluation history with scores	Exports to CSV
Administration	Users, Departments, Sites, Audit logs, Notifications	Manage users and roles, departments with head and deputy, and sites; read the audit trail	The administrator cannot carry out business steps

SECTION 09

Supplier lifecycle

A supplier can be asked for a price while its checks are in progress, because a quote commits nothing. It cannot win an award or receive a purchase order until its KYS is complete, a second person has approved its activation and it is marked spend-authorised.

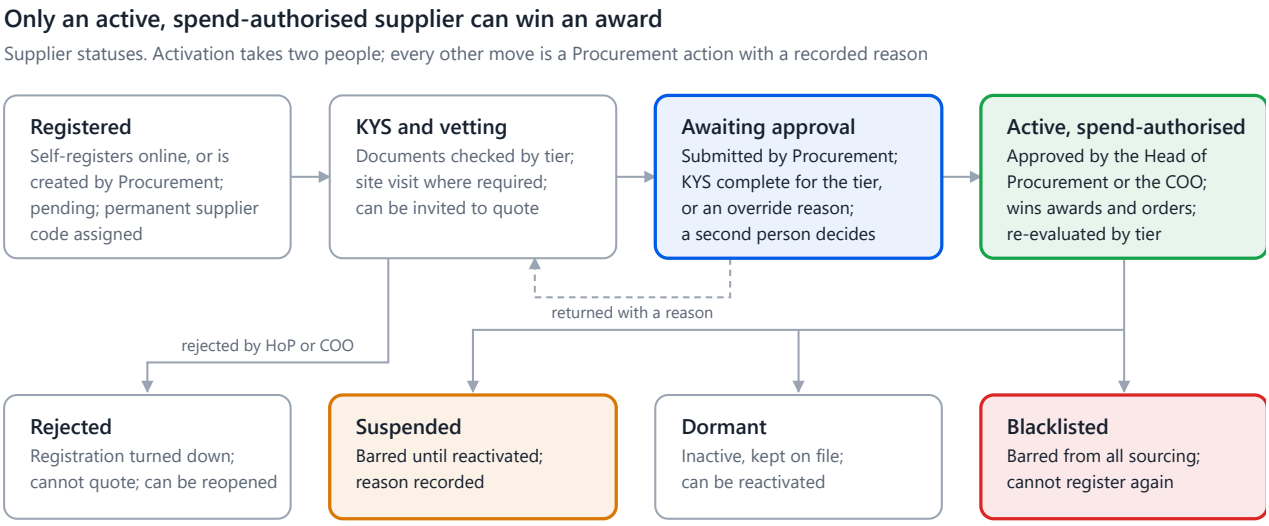


Figure 2. Supplier statuses. Activation takes two people: Procurement submits, and the Head of Procurement or the COO approves. A returned request leaves the supplier pending; a rejected registration can be reopened; suspended and dormant suppliers can be reactivated. Every change is logged with who made it and why.

Two-person activation

Activation is what gives a supplier the right to be paid, so since 5 November 2026 no one person can grant it. The request is kept on the supplier record while it is open, and the supplier stays pending, so it can still be invited to quote.

Step	Who	What happens
Submit	Procurement officer or Head of Procurement	Assigns the tier, works through the KYS checklist and submits, with notes. If the checklist is short of what the tier requires, a KYS override reason is mandatory and travels with the request. The Heads of Procurement and COOs, other than the submitter, are notified
Approve	Head of Procurement or COO, never the submitter	The supplier becomes active and spend-authorised. An override is recorded as granted by the approver, who accepted it. If the checklist changed while the request waited and is no longer complete, approval is refused until it is returned and resubmitted
Return	Head of Procurement or COO	With a reason. The supplier stays pending and the submitter is told; it can be resubmitted once the gap is closed

Step	Who	What happens
Reject registration	Head of Procurement or COO	With a reason. The supplier becomes rejected, is told why, and cannot quote, be awarded or be ordered from. Unlike blacklisting, it says nothing against the company
Reopen registration	Head of Procurement or COO	With a reason. The supplier returns to pending and activation again takes two people
Reactivate	Procurement	A suspended or dormant supplier is reactivated by Procurement with a reason, as before. This is not onboarding, so a second approval is not required

Every submission, approval, return, rejection and reopening is written to the audit log and to the supplier's own status history with the reason given. The older verify and approve actions on a pending supplier now submit it rather than activate it, so an older screen or script cannot activate a supplier alone. The dashboard, the supplier list filter *Awaiting my approval* and the sidebar badge all show an approver the requests that are theirs to decide.

Due diligence by tier

The depth of the Know Your Supplier (KYS) check depends on the supplier's risk tier. An unclassified supplier gets the full check, so low risk is never assumed by default.

Tier	Meaning	KYS items required	Re-evaluated every
Critical	Sole source, or business stops if they fail	All 16 required items	3 months
Strategic	High spend or long-term partner	All 16 required items	6 months
Tactical	Routine, bought competitively	5: CR14, CR6, tax clearance, NSSA, insurance	12 months
Transactional	Low value, low risk, easily replaced	2: CR14, tax clearance	24 months
Unclassified	Not yet assessed	All 16 required items	3 months

The full checklist has 20 items in seven groups: registration, tax, statutory compliance, licences, company capability, financial standing, and health, safety and environment.

Ongoing controls

- **Registration.** A company registers once. The registration number is compared ignoring case, spaces and hyphens; a second registration is refused, and a blacklisted company gets a neutral refusal that does not disclose why. A contact phone number is required, and a registration that fails part-way leaves no half-made login behind.
- **Identity.** Each supplier gets a permanent code on first save, such as `DOKUMA-SUP-00042`, which never changes even if the name or registration number is corrected. Business type (manufacturer, distributor, wholesaler, retailer, service provider, contractor, other) and entity type (company or individual) are optional and validated wherever a supplier is created or imported.

- **Document expiry.** Uploaded documents carry expiry dates. Reminders go out 60, 30 and 7 days before expiry. A required document that has passed its expiry date blocks awards and orders on its own, whether or not the hourly job has run.
- **Vetting.** Procurement records site visits (date, who and notes), the vetting result, and how the supplier was found (newspaper advert, direct application, referral).
- **Bank details.** A change is requested by the supplier, checked by a call-back to a phone number that was on file before the request, and approved by a third person. The phone numbers are frozen while a change is open, so a number added with the change can never be the one called. A staff change to a supplier's sign-in email is told to both the old and the new address.
- **Evaluation.** After delivery, Procurement scores the supplier on 7 criteria: credit terms, contracts in place, market reputation, pricing, delivery, ease of dealing and consistent quality.
- **Ranking of quotations.** Responses to an RFQ are scored out of 100: price 50, delivery 20, past performance 20, completeness 10. The ranking is advisory, and a snapshot is saved when the RFQ closes.

SECTION 10

Where and how: architecture and technology

Procureline is a TypeScript web application hosted on Vercel, with its data in MongoDB Atlas. Each business unit has its own web address and its own database, and a session opened in one unit is refused by every other.

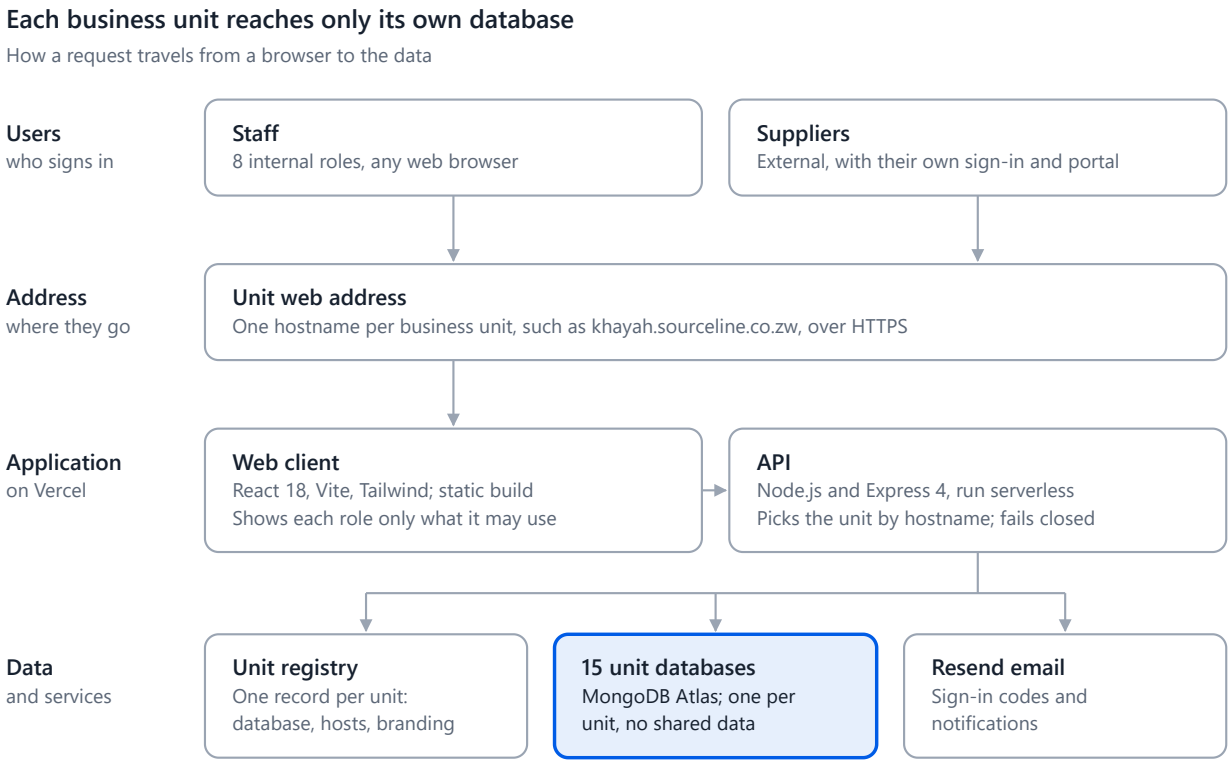


Figure 3. Architecture. The API works out the business unit from the web address of each request; if it cannot, it refuses the request rather than read whichever database happens to be open.

Technology

Layer	Technology
Web client	React 18, Vite 5, React Router 7, Tailwind CSS 3
API	Node.js, Express 4, TypeScript 5.5
Shared rules	One package of statuses, roles, thresholds and KYS rules, used by both client and API
Database	MongoDB Atlas, Mongoose 8; 26 data models
Sign-in	Password (bcrypt hash), then a 6-digit code by email; JSON Web Tokens
Email	Resend

Layer	Technology
Documents	PDFKit for printed purchase orders and issue notes, the same PDF a supplier downloads; SheetJS for bulk supplier and stock import; supplier compliance files (up to 5 MB each) stored on the supplier record and served one at a time
Hosting	Vercel: the client as a static site, the API as serverless functions
Scheduled jobs	An hourly in-process timer on a long-running server. On Vercel there is no long-lived process, so an external scheduler calls a secret-protected internal endpoint that runs the same hourly pass for every business unit
Exchange rates	Stored rates to US dollars per currency, used for the COO limit and group rules

SECTION 11

Security, audit trail and notifications

Every user signs in with two factors, every request is checked against the user's role and business unit, and every material action is recorded. Controls are enforced on the server, so hiding a button is never the only protection.

Access control

Control	How it works
Sign-in	Email and password, then a 6-digit code sent by email. The code expires after 10 minutes and allows 5 attempts
Passwords	Stored only as bcrypt hashes; users reset their own through an emailed link
Sessions	Signed tokens that expire after 7 days and are bound to one business unit
Supplier access	Separate sign-in route; suppliers never reach internal screens or other suppliers' bids
Role checks	Every API route checks the role and, where relevant, the department or site. The administrator can read business records but not change them
Separation of units	Each unit has its own database; a request without a known unit is refused
Sealed bids	Quotations cannot be compared until the RFQ closes
Bank details	Masked on screen, including in the change history; changes need a call-back to a number on file before the request and an independent approver
Supplier documents	Served one file at a time to staff who may read suppliers and to the supplier that owns them; never cached, and with headers that stop a browser running an uploaded file
Supplier portal data	A supplier sees its own orders, its own acknowledgement and its own invoices with payment history; internal approvals, match workings and override reasons are left out

Control	How it works
Two-person rules	Awards, bank-detail changes and supplier activation each need two different people; the server refuses the same person on both sides

Audit trail

Every create, update, delete, approval, rejection, submission, upload, download, status change, sign-in and failed sign-in is logged. Each entry records the user, their role, the record, the time, the IP address and the browser. Supplier activation submissions, approvals, returns, rejections and reopenings are logged with the reason and, for an override, the reason the approver accepted. Administrators can search the log, and an evidence pack endpoint gathers proof of controls for auditors.

Data retention

Record	Kept for
Orders, invoices, payments, GRVs, quotations, RFQs, requisitions	7 years from the end of the financial year they closed
Audit trail and bank-change records	7 years
Active supplier profiles	While the relationship lasts
Dormant or never-approved supplier profiles	2 years without activity (7 years if they ever traded)
Staff accounts	2 years after deactivation
In-app notifications	12 months if read, 24 months if unread; deleted automatically
Sign-in codes	1 day after expiry; deleted automatically

The retention policy is a draft awaiting approval by Finance and Legal.

Notifications

Users get in-app notifications, and email through Resend, when something reaches their step. Examples: a requisition to approve, a PO at their approval stage, a quotation selected for authorisation, a supplier awaiting activation approval, an activation approved or returned, a registration rejected or reopened, an RFQ closing with fewer than 3 quotes, an over-budget approval, an invoice to match, low stock, expiring supplier documents and suppliers due for review. Suppliers are told when an RFQ they are invited to opens, what stores accepted and rejected on each GRV, and the outcome of their registration; the buyer is told when the supplier acknowledges an order.

SECTION 12

Compliance, rollout, assumptions and risks

The software is built and proven end to end. What stands between it and live use is infrastructure and set-up owned by the group, not further development.

Compliance with the procurement procedure

The system is built to Tefoma's Central Procurement Procedure, Rev 9.0 (25 November 2025). `PROCEDURE_COMPLIANCE.md` maps each clause to where the code enforces it.

Clause	Procedure says	System does	Status
5.2.1	Department head selects the winning quotation	Procurement selects with a justification; the Head of Procurement authorises	Deliberate departure, decided 28 Oct 2026
6.3.12	Order approved by Procurement, department head, Finance, then COO	Head of Procurement, Finance, then COO; the department head step was removed	Deliberate departure, decided 28 Oct 2026; matrix still to be updated
5.5, 6.2.1	Supplier site visits; newspaper onboarding	Site visits, vetting result and onboarding method recorded	Added 29 Oct 2026
6.6.3, 6.6.7–6.6.9, 6.8.5	MSDS for hazardous goods; return of faulty goods; witnesses at receipt	MSDS reference, 5-day return deadline and witness names on the GRV	Added 29 Oct 2026
6.2.2, 6.2.5	Onboarding framework; approved suppliers list	A supplier joins the approved list only when a second person, the Head of Procurement or the COO, approves its activation	Added 5 Nov 2026
6.3.14	Order emailed to the supplier	Supplier is notified, sees the order in the portal and can download the order PDF; no PDF attached to the email	Partial
6.6.2	Quality check by department representatives	Condition recorded per line; no department sign-off	Partial

Rollout plan

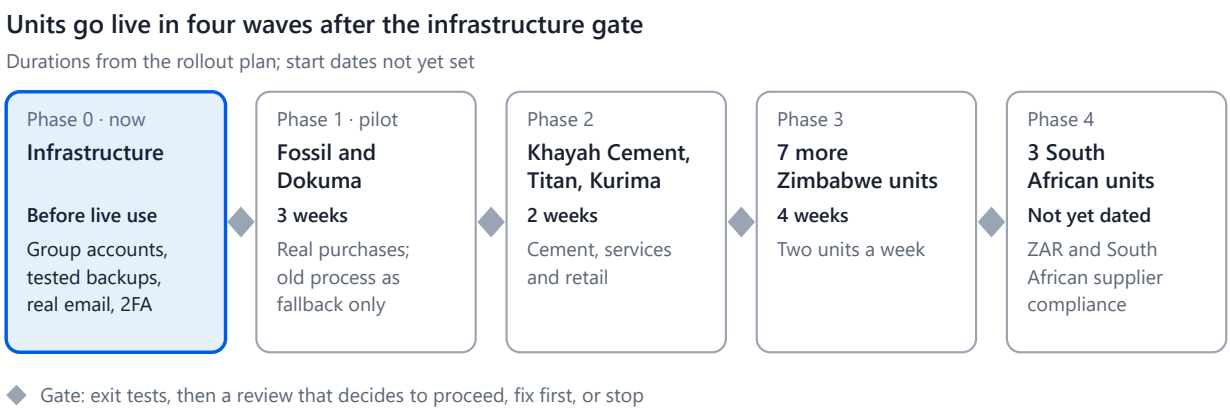


Figure 4. Rollout plan. Phase 0's exit test: an administrator resets their password from a real email and signs in with a real emailed code, on a group-owned address. Phase 1's exit tests include ten real requisitions paid, one order above the COO limit through every approval, and stock agreeing with a physical count.

Assumptions

- Every user, including each supplier contact, has an email address they can reach, because sign-in depends on an emailed code.
- Stores keeps inventory in the system, not in parallel spreadsheets.
- Each unit's COO limit and approval chain match its real delegation of authority. This must be confirmed per unit before go-live.
- The group owns the source code, the data and every service account.
- Each unit has at least one Head of Procurement or COO user, otherwise no supplier can be activated there.

Constraints

- Where the procedure and convenience conflict, the procedure governs, unless the business records a deliberate departure.
- Sites have variable connectivity, so every screen must work in a mobile browser, approvals especially.
- Suppliers are external, with no access to the company network and mixed technical skills.

Risks and open items

Risk	Effect if left	Action
Development, staging and production share one database cluster and one database user	Development credentials can change live data; testing load reaches production	Give each environment its own cluster or user
Database backup tier not confirmed	Payment records could be lost with no restore	Move to a paid tier and time a real restore
Hosting, database and email accounts not yet in the group's name	One person is a single point of failure	Complete the handover checklist in the rollout plan

Risk	Effect if left	Action
Real email delivery not yet verified for administrators	Nobody can sign in with real codes; password resets fail	Verify the sending domain; confirm codes arrive
Database cluster limit of 500 collections; 197 in use after the 3 Nov 2026 clean-up	Bringing all units live could hit the limit	Check capacity with `sbu:capacity` before each wave
Email links point to one fixed web address	A link may open the wrong unit	Send per-unit links
Older documents and test packs describe the four-stage PO chain	Testers and auditors check against the wrong rule	Update the guides, the test packs and the compliance matrix
Orders left at the old department-head step	They wait for the wrong person	The Head of Procurement can now approve them
The hourly jobs do not run by themselves on the hosted API, because serverless functions have no timer	Document-expiry, low-stock, RFQ-deadline and re-evaluation alerts and the retention job never fire in production	Set `CRON_SECRET` and schedule a call to `/api/internal/run-alert-jobs` every hour (Vercel Cron or another scheduler)
Suppliers created before 5 Nov 2026 have no permanent code	They cannot be found by code; the register is inconsistent	Run `backfill:supplier-codes` for each unit, dry run first, then with `--apply`
Older guides and test packs describe single-person supplier activation	Testers expect a pending supplier to go active on verify	Update the guides and test packs to the submit-then-approve flow in section 9

SECTION 13

Glossary

Term	Meaning
Activation request	A pending supplier put forward by Procurement for a second person to approve. Open until approved, returned or rejected
Award	The decision to buy from a supplier's quotation, made in full or line by line
Business unit (SBU)	One of the 15 group companies. Each has its own address, database and settings
COO limit	The order value in US dollars at or above which the COO must approve. The default is USD 5,000
CR14 / CR6	Zimbabwean company registration forms listing directors (CR14) and the registered address (CR6)
Deputy head	A user who can act for a department head on that department's requisitions
GRV	Goods received voucher: the stores record of what physically arrived

Term	Meaning
Head of Procurement (HoP)	The department head of the Procurement department. Authorises awards and gives the first approval on orders
KYS	Know Your Supplier: the due diligence checklist a supplier must pass before it can be paid
MSDS	Material safety data sheet, required for hazardous goods
NSSA	National Social Security Authority (Zimbabwe); its compliance certificate is a KYS item
PO	Purchase order: the commitment to buy, sent to the supplier after approval
Requisition (PR)	A request from a department for goods or services
Rejected (registration)	A supplier registration turned down with a reason. The company cannot quote or be awarded, and the registration can be reopened. Not a blacklisting
RFQ	Request for quotation, sent to invited suppliers
Sealed bid	A quotation no one can compare with others until the RFQ closes
Spend-authorised	A supplier whose activation has been approved and who may receive awards and orders
Supplier code	The permanent register number given to a supplier on first save, such as DOKUMA-SUP-00042
Three-way match	Comparing the order, the goods received and the invoice before payment
Tier	A supplier's risk class (critical, strategic, tactical, transactional, unclassified), which sets the depth of its KYS and how often it is reviewed
Two-person rule	A control that needs two different people, one to propose and one to approve: awards, bank-detail changes and supplier activation
Waiver	Approval to award with fewer than three quotations, with a reason from a fixed list